CliniClaimZA-LLM Data Card

Dataset Documentation & Ethical Governance Card for South African Healthcare Claim Adjudication & Triage

1. Input versus Output

- **User & System Inputs (Text):**
  - **Task 1 (Scheme Adjudication):** Structured free-text clinical claim records containing Medical Specialty and Clinical Description.
  - **Task 2 (Conversational Intake):** Free-text patient inquiries and symptom descriptions using natural layperson language and local South African idioms, paired with medical scheme context (such as scheme name, plan tier, or provider network status).
- **Model Outputs & Predicted Labels:**
  - **Task 1 (Classification Labels):** Explicit policy determinations categorized into one of three standardized adjudication classes: *In-Cover*, *Out-of-Cover*, or *Requires Pre-Auth*, along with relevant treatment or prescription context.
  - **Task 2 (Conversational Triage):** Non-diagnostic conversational triage guidance, structured benefit explanations, and actionable next steps detailing statutory Prescribed Minimum Benefits (PMBs), Designated Service Provider (DSP) network rules, and pre-authorization requirements—without making unauthorized clinical diagnoses or speculating on out-of-pocket financial costs.

2. Classification Problem

- **What was the classification task?** The classification task performs multi-class administrative claim adjudication and benefit triage. It evaluates clinical encounter descriptions, proposed medical treatments/prescriptions, and medical specialties against South African medical scheme regulations and policy rules.
- **What categories or classes were you working with, and why?** The model classifies inputs into three standardized, mutually exclusive policy decision outcomes:
  1. **In-Cover:** Healthcare services fully funded by the scheme, including statutory Prescribed Minimum Benefits (PMBs) and Designated Service Provider (DSP) network care.
  2. **Out-of-Cover:** Non-funded, excluded, or out-of-pocket expenses (such as non-PMB elective procedures or unapproved out-of-network care).
  3. **Requires Pre-Auth:** Conditional coverage that requires prior administrative authorization, scheme pre-approval, or clinical motivation before treatment can proceed.
- **Why these categories exist:** These categories mirror the real-world administrative workflow of South African medical schemes. By restricting predictions to administrative coverage status rather than clinical severity or diagnosis, the classification task eliminates unexpected out-of-pocket medical expenses and prevents AI hallucination or unauthorized medical diagnoses.

3. Labeling

- **Where did your labels come from?** Labels were established by mapping clinical encounter descriptions and proposed treatments to standard South African medical scheme administrative rule sets—categorized into *In-Cover*, *Out-of-Cover*, and *Requires Pre-Auth*. Prompts and responses reflect statutory Prescribed Minimum Benefits (PMBs) and pre-authorization requirements across major scheme plan structures.

- **How did you check for consistency and accuracy?**

  1. **Programmatic Balance Verification:** Ensured strict class balance across all three categories in both Task 1 (98 samples per class) and Task 2 (60 samples per class).

  2. **Automated Data Cleaning:** Applied text-cleaning pipelines to strip HTML artifacts, normalize unicode characters, and standardize whitespace across all records.

  3. **No Expert Review Disclaimer:** Data engineering and formatting were performed programmatically; the dataset did not undergo manual auditing or verification by human medical domain experts.

4. Data Creation

- **How was the data collected?** The dataset was constructed by combining open-source clinical encounter and medical dialogue datasets from HuggingFace and Kaggle with public South African healthcare regulatory frameworks. Raw medical descriptions, specialties, and dialogue pairs were mapped to statutory guidelines from the Council for Medical Schemes (CMS)—including Prescribed Minimum Benefit (PMB) definitions, Chronic Disease Lists (CDLs), and Diagnosis Treatment Pairs (DTPs)—alongside annual scheme rules and Designated Service Provider (DSP) network guidelines from major South African insurers.

- **Direct Sources vs. Localized Synthetic Data in Low-Resource Settings:** The dataset pairs direct regulatory rule sets for administrative accuracy with localized synthetic scenario generation using Gemini Notebook (NotebookLM) to model layperson South African English phrasing and local idioms. Synthetic prompt-response engineering addresses low-resource healthcare navigation challenges in South Africa, where complex clinical terminology often lacks direct 1:1 translations across indigenous language contexts. Prompts map layperson symptom descriptions to structured policy determinations and plain-language triage, helping patients articulate treatment needs during doctor consultations.

- **Data Engineering & Quality Assurance Note:** Data ingestion, transformation, and split formatting were executed via automated data engineering pipelines using regulatory logic templates without manual crowd-worker review.

5. Web Collection Ethics

- **Do you plan to source data from the web? If yes, which sites or platforms?**

  – **Current Scope:** Sourced from existing de-identified public datasets on HuggingFace and Kaggle, alongside medical aid treatments, clinical descriptions, and policy decision prompts synthesized using Gemini Notebook (NotebookLM) to create a clean, balanced baseline.

  – **Future Scope:** Future work will directly scrape and index real annual policy documents, benefit brochures, Prescribed Minimum Benefit (PMB) codes, and Designated Service Provider (DSP) network rosters from official South African medical scheme platforms—specifically market-leading schemes like Discovery Health and Clientèle, which broadly represent the South African workforce—alongside the Council for Medical Schemes (CMS) regulatory portal.

- **What tools will you use—AI tools, human workers, or both?** Data formatting and synthesis relied on AI generation tools (Gemini Notebook) and automated Python data-cleaning pipelines. No human crowd-workers or manual auditing teams were employed.

- **Have you addressed issues of consent, copyright, and privacy?**

  – **Privacy & POPIA Compliance:** Aligned strictly with South Africa's Protection of Personal Information Act (POPIA). All inputs rely on de-identified open-source data from Kaggle and HuggingFace and synthesized prompts. No Personal Health Information (PHI), real patient records, or private claim logs were collected or used.

  – **Copyright, Licensing & Academic Fair Use:** Sourced datasets from Kaggle and HuggingFace were reused under their respective open-source licenses strictly for non-commercial academic research and capstone development within the Google DeepMind AI Course. Public-facing scheme guidelines and statutory PMB rules are used for

analytical indexing and instruction tuning under fair-use research standards without reproducing full proprietary texts.

6. Bias and Representation

- **Who might be left out of your dataset?**
  - **Socio-Economic Exclusions:** Uninsured and unemployed populations operating outside the private medical scheme ecosystem, who rely exclusively on public healthcare services.
  - **Accessibility & Language Barriers:** Individuals with literacy constraints or acute trauma, as well as non-English speakers whose health concepts and local idioms are underrepresented in standard English clinical corpora.
  - **Model Scope Limitations:** Members of smaller, specialized medical schemes or niche plan tiers whose specific coverage exemptions differ from major market standards.
- **How do you ensure the dataset reflects community diversity and underrepresented voices?**
  - **Public & Private Pathway Integration:** Incorporates public healthcare triage directives alongside private scheme rules, ensuring that emergency cases or non-covered queries provide clear administrative guidance to statutory Prescribed Minimum Benefits (PMBs) or public healthcare services regardless of financial status.
  - **Plain-Language Conversational Intake:** Maps free-text layperson symptom descriptions directly to clear, plain-language administrative outcomes and conversational triage guidance without requiring users to know complex medical jargon or scheme terminology.
  - **Class & Triage Parity:** Maintains strict class balance (*In-Cover*, *Out-of-Cover*, *Requires Pre-Auth*) and equal representation across triage categories to prevent algorithmic bias toward automatic denial or over-authorization.

7. Values and Community Alignment

- **How does your dataset design reflect the values you identified?**
  - **Environmental Responsibility & Resource Efficiency:** The dataset utilizes a standardized, token-optimized instruction schema ({"input": ..., "output": ...}) that minimizes computational overhead during LLM fine-tuning and inference, reducing carbon and energy footprints.
  - **Psychological Safety & Medical Integrity:** Eliminates alarming or speculative diagnostic outputs, replacing sycophantic AI tendencies with gentle, non-diagnostic premise checking and administrative clarity regarding coverage rules.
  - **Transparency & Empathetic Triage:** Structures conversational intake and classification predictions into clear, non-prescriptive administrative guidance that respects patient agency while preventing health anxiety.
- **How will your community benefit from this dataset, and how will you ensure accountability?**
  - **Community Benefits:** South African healthcare consumers and scheme members benefit from fast, transparent benefit-matching and conversational triage. This mitigates unexpected out-of-pocket financial risks, demystifies complex scheme rulebooks, and helps patients articulate their treatment needs clearly during clinical consultations.

  1. **POPIA Compliance & Privacy:** Strict adherence to South Africa's Protection of Personal Information Act through de-identified dataset reuse and de-identified synthetic prompt engineering.
  2. **Open-Source Evaluation Benchmark:** Establishes a non-sycophantic, reproducible evaluation benchmark for South African medical AI research.
  3. **Dynamic Policy Adaptability:** Designed to allow annual data updates as statutory Prescribed Minimum Benefits (PMBs) and medical scheme rulebooks evolve, ensuring ongoing accuracy for scheme members.